

Large amplitude solution of BGK model

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Outline

- 1 Boltzmann - BGK model
- 2 Large amplitude solution of BGK model

Boltzmann - BGK model

Boltzmann Equation (Ludwig Boltzmann (1872))

- Boltzmann equation

$$\partial_t F + \underbrace{v \cdot \nabla_x F}_{\text{transport}} = \underbrace{Q(F, F)}_{\text{collision}},$$

where

$$Q(F, F) = \int_{\mathbb{R}^3} \int_{\mathbb{S}^2} q(\omega, |v - v_*|) (F(v'_*)F(v') - F(v_*)F(v)) d\omega dv_*,$$

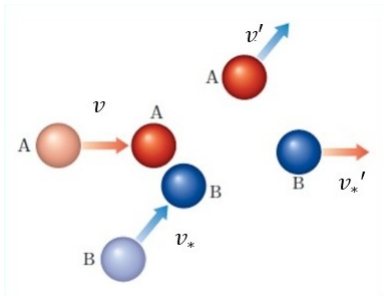


Figure: Elastic collision

Boltzmann BGK model (Bhatnagar-Gross-Krook (1954))

- Relaxation operator $Q(F, F) \rightarrow \rho^a(\mathcal{M}(F) - F)$:

$$\partial_t F + v \cdot \nabla_x F = \rho^a(\mathcal{M}(F) - F), \quad a > 0,$$

where $\mathcal{M}(F)$ is given by

$$\mathcal{M}(F) = \frac{\rho(x, t)}{\sqrt{2\pi T(x, t)}^3} \exp\left(-\frac{|v - U(x, t)|^2}{2T(x, t)}\right).$$

- Density, bulk velocity, temperature:

$$\rho(x, t) = \int_{\mathbb{R}^3} F(x, v, t) dv,$$

$$\rho(x, t)U(x, t) = \int_{\mathbb{R}^3} F(x, v, t) v dv,$$

$$3\rho(x, t)T(x, t) = \int_{\mathbb{R}^3} F(x, v, t) |v - U(x, t)|^2 dv.$$

Conservation laws and H -theorem

- Conservation law

$$\frac{d}{dt} \int_{\Omega} \int_{\mathbb{R}^3} F(1, v, |v|^2) dv dx = 0.$$

- H -Theorem

$$\frac{d}{dt} \mathcal{H}(t) = \frac{d}{dt} \int_{\Omega} \int_{\mathbb{R}^3} F \ln F dv dx \leq 0.$$

- Local equilibrium

$$\mathcal{M}(F) = \frac{\rho}{\sqrt{2\pi T}^3} e^{-\frac{|v-U|^2}{2T}}.$$

Property of the Boltzmann and BGK model

- Expand $F = \mu + \sqrt{\mu}f$ around global Maxwellian:

$$\mu(v) = \frac{1}{\sqrt{2\pi}^3} e^{-\frac{|v|^2}{2}}.$$

- Boltzmann equation near global Maxwellian:

$$\partial_t f + v \cdot \nabla_x f = Lf + \Gamma(f, f).$$

- BGK model near global Maxwellian:

$$\partial_t f + v \cdot \nabla_x f = Pf - f + \Gamma(f).$$

Existence of the Boltzmann equation

- Renormalized solution [1989 Diperna, Lion. Annals]
- $H_{x,v}^2$ small: [2002,2003 Yan Guo. CPAM, Invent]
- Pointwise behavior: [2004 Tai-Ping Liu, Shih-Hsien Yu. CPAM]
- $L_{x,v}^{\infty,q}$ small: [2010 Yan Guo. ARMA]
- $L_x^1 L_v^\infty$ small with small relative entropy [Duan, Huang, Wang, and Yang 2017. ARMA]
- L^2 small L^∞ large [Duan, Wang 2019. Adv Math]

Existence of the BGK model

- Weak sol [1989, Perthame, JDE]
- Weak sol [1993, Perthame, Pulvirenti. ARMA]
- $H_x^2 L_v^2$ small [2010 Yun, JMP]
- L^2 small L^∞ large with additional assumption that (ρ, U, T) is close to $(1, 0, 1)$. [2017 Duan, Wang, Yang: Sci. Sin. Math.]

Large amplitude solution of BGK model

Main Theorem

Theorem

Let $F_0 = \mu + f_0$ is non-negative, and satisfies

$$\inf_{(t,x) \in [0,\infty) \times \mathbb{T}^3} \int_{\mathbb{R}^3} F_0(x - vt, v) dv \geq C_0,$$

for some positive constant $C_0 > 0$. Then, for any $M_0 > 0$, there exists $\varepsilon = \varepsilon(M_0, C_0)$ such that if initial data f_0 satisfies ($q > 10$)

$$\|f_0\|_{L_{x,v}^{\infty,q}} \leq M_0, \quad \mathcal{E}(F_0) = \int_{\mathbb{T}^3 \times \mathbb{R}^3} (F_0 \ln F_0 - \mu \ln \mu) dv dx \leq \varepsilon,$$

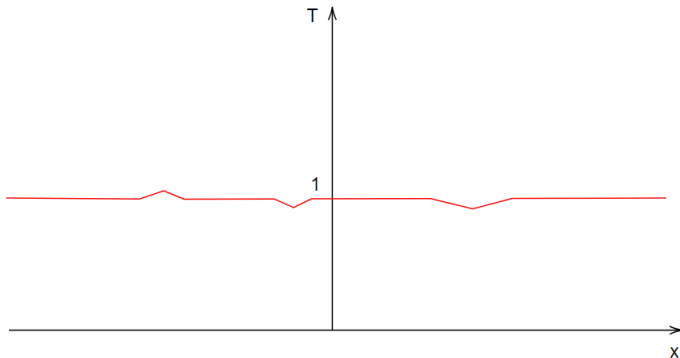
then there exists a unique global-in-time solution to the BGK model. Moreover f satisfies

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \leq C_{M_0} e^{-kt},$$

where C_{M_0} depending on M_0 , and k are positive constants.

What is large amplitude?

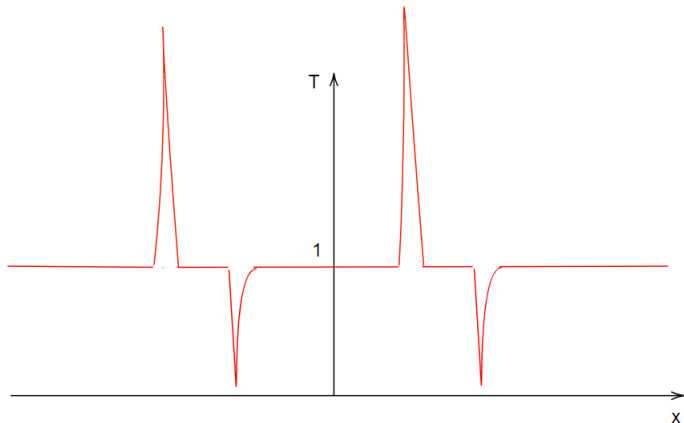
- $H_{x,v}^2$ small or $L_{x,v}^{\infty,q}$ small initial data



What is large amplitude?

- Consider $F_0 = \frac{1}{\sqrt{2\pi T_0^3}} e^{-\frac{|v|^2}{2T_0}}$. Then we have

$$\mathcal{E}(F_0) = \int_{\mathbb{T}^3 \times \mathbb{R}^3} (F_0 \ln F_0 - \mu \ln \mu) dv dx = \int_{\mathbb{T}^3} -\frac{3}{2} \ln T_0 dx \leq \varepsilon,$$



Difficulty

- L^∞ estimate of the Boltzmann equation

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \lesssim C \|f_0\|_{L_{x,v}^{\infty,q}} \left(1 + \int_0^t \|f(s)\|_{L_{x,v}^{\infty,q}} ds \right) e^{-\lambda t} + D, \quad D \ll 1.$$

- L^∞ estimate of the BGK model

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \lesssim C (\|f_0\|_{L_{x,v}^{\infty,q}}) \left(1 + \int_0^t \left[\|f(s)\|_{L_{x,v}^{\infty,q}} \right]^n ds \right) e^{-\lambda t} + D.$$

Why $F = \mu + f$?

- We define

$$\int_{\mathbb{R}^3} F \left(1, v, \frac{|v|^2 - 3}{\sqrt{6}} \right) dv = (\rho, \rho U, G).$$

- Global Maxwellian of $(\rho, \rho U, G)$ is equal to $(1, 0, 0)$.
- Make convex combination:

$$\rho_\theta = \theta \rho + (1 - \theta), \quad \rho_\theta U_\theta = \theta \rho U, \quad G_\theta = \theta G.$$

Why $F = \mu + f$?

- Nonlinear term of BE:

$$\Gamma(f, f) = \frac{1}{\sqrt{\mu}} Q(\sqrt{\mu}f, \sqrt{\mu}f).$$

- Nonlinear term of BGK:

$$\begin{aligned} \Gamma(f) &= \sum_{1 \leq i, j \leq 5} \int_0^1 \left[\nabla_{(\rho_\theta, \rho_\theta U_\theta, G_\theta)}^2 \mathcal{M}(\theta) \right]_{ij} \frac{1}{\sqrt{\mu}} (1 - \theta) d\theta \\ &\quad \times \int_{\mathbb{R}^3} f e_i \sqrt{\mu} dv \int_{\mathbb{R}^3} f e_j \sqrt{\mu} dv. \end{aligned}$$

where $(e_1, \dots, e_5) = (1, v, (|v|^2 - 3)/\sqrt{6})$.

- Unavailable of weighted L^∞ estimate. $\rightarrow$ We use $F = \mu + f$.

Mild form

- Linearization with respect to $F = \mu + f$:

$$\partial_t f + v \cdot \nabla_x f + f = \mathbf{P}f + \Gamma(f),$$

where

$$\begin{aligned} \mathbf{P}f &= \int_{\mathbb{R}^3} f dv \mu + \int_{\mathbb{R}^3} f v dv \cdot (v \mu) + \int_{\mathbb{R}^3} f \frac{|v|^2 - 3}{\sqrt{6}} dv \left(\frac{|v|^2 - 3}{\sqrt{6}} \mu \right), \\ \Gamma(f) &= \sum_{1 \leq i, j \leq 5} \int_0^1 \left[\nabla_{(\rho_\theta, \rho_\theta U_\theta, G_\theta)}^2 \mathcal{M}(\theta) \right]_{ij} (1 - \theta) d\theta \int_{\mathbb{R}^3} f e_i dv \int_{\mathbb{R}^3} f e_j dv. \end{aligned}$$

- Mild form:

$$\begin{aligned} f(t, x, v) &= e^{-t} f_0(x - vt, v) \\ &\quad + \int_0^t e^{-(t-s)} (\mathbf{P}f + \Gamma(f))(s, x - v(t-s), v) ds. \end{aligned}$$

Sketch of Proof

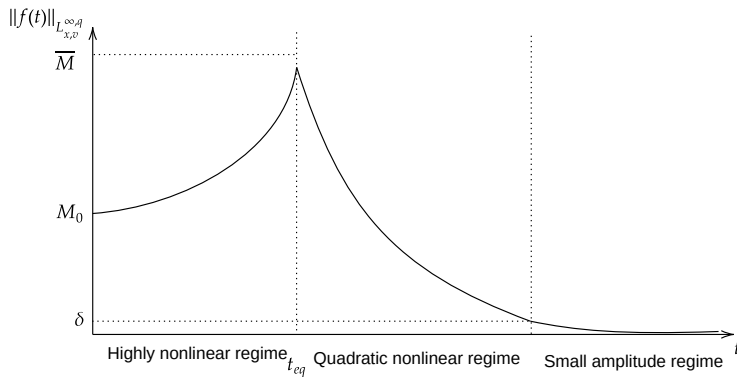


Figure: Three different regimes

Sketch of Proof

- 1 If $\sup_{0 \leq s \leq t_*} \|f(s)\|_{L_{x,v}^{\infty,q}} \leq \overline{M}$, then we have t_{eq} such that $|(\rho, U, T) - (1, 0, 1)| \leq \delta$ for $t \in [t_{eq}, \infty]$. We can reduce the nonlinear term after t_{eq} .
- 2 Estimate growth of the nonlinear term until t_{eq} .
- 3 L^∞ decaying estimate. Choosing $\overline{M}$, we can close the argument.
- 4 Smallness problem

Sketch of Proof (Step 1)

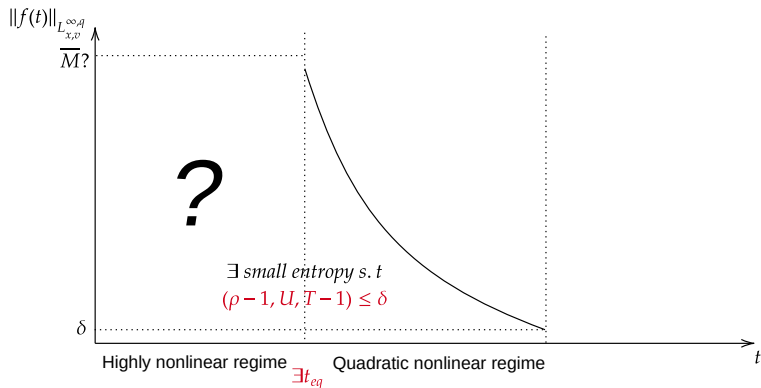


Figure: Step 1

Sketch of Proof of Step 1

Lemma

Under a priori assumption

$$\sup_{0 \leq s \leq t_*} \|f(s)\|_{L_{x,v}^{\infty,q}} \leq \bar{M}, \quad \text{for } q > 10,$$

there exists $\epsilon_{(\bar{M}, t_, \delta)}$ such that if $\mathcal{E}(F_0) \leq \epsilon_{(\bar{M}, t_*, \delta)}$, then the following estimate holds on $t \in [0, t_*]$:*

$$\left| \int_{\mathbb{R}^3} \langle v \rangle^2 f(t, x, v) dv \right| \leq C_q M_0 e^{-t} + \frac{3}{4} \delta.$$

Sketch of Proof of Step 1

- Divide the velocity region and put the mild form:

$$\begin{aligned} \left| \int_{\mathbb{R}^3} \langle v \rangle^2 f(t, x, v) dv \right| &\leq \frac{C_q}{N^{q-5}} \overline{M} + e^{-t} \int_{|v| \leq N} \langle v \rangle^2 |f_0(x - vt, v)| dv \\ &+ \int_0^t e^{-(t-s)} \int_{|v| \leq N} \langle v \rangle^2 |\mathbf{P}f(s, x - v(t-s), v)| dv ds \\ &+ \int_0^t e^{-(t-s)} \int_{|v| \leq N} \langle v \rangle^2 |\Gamma(f)(s, x - v(t-s), v)| dv ds. \end{aligned}$$

for $t \in [0, t_*]$ and arbitrary $N > 0$.

Sketch of Proof of Step 1

We consider $\mathbf{P}f$ part:

$$\int_0^t \int_{|v| \leq N} e^{-(t-s)} \langle v \rangle^2 |\mathbf{P}f(s, x - v(t-s), v)| dv ds$$

Divide the integral region

$$I = \int_{t-\lambda}^t \int_{|v| \leq N} \int_{\mathbb{R}^3} \cdots dudvds,$$

$$II = \int_0^{t-\lambda} \int_{|v| \leq N} \int_{|u| \geq 2N} \cdots dvds,$$

$$III = \int_0^{t-\lambda} \int_{|v| \leq N} \int_{|u| \leq 2N} \cdots dudvds.$$

Sketch of Proof of Step 1

Estimate for III : Using $\langle v \rangle^2 \leq N^2$ and $(1 + |u| + |u|^2) \leq 4N^2$, we have

$$III \leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \int_{|v| \leq N} \int_{|u| \leq 2N} |f(s, x - v(t-s), u)| du dv ds.$$

Change of variable $y = x - v(t-s)$ with $dy = -(t-s)^3 dv$:

$$III \leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \frac{1 + (N(t-s))^3}{(t-s)^3} \int_{\mathbb{T}^3} \int_{|u| \leq 2N} |f(s, y, u)| du dy ds.$$

Sketch of Proof of Step 1

Divide the integral into $\mathbf{1}_{|f(s,y,u)|>\mu(u)}$ and $\mathbf{1}_{|f(s,y,u)|\leq\mu(u)}$:

$$\begin{aligned} III &\leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \frac{1 + (N(t-s))^3}{(t-s)^3} \\ &\quad \times \left(\int_{\mathbb{T}^3} \int_{|u|\leq 2N} |f(s,y,u)| \mathbf{1}_{|f(s,y,u)|>\mu(u)} du dy \right. \\ &\quad \left. + \int_{\mathbb{T}^3} \int_{|u|\leq 2N} \frac{1}{\sqrt{\mu(u)}} |f(s,y,u)| \mathbf{1}_{|f(s,y,u)|\leq\mu(u)} du dy \right) ds. \end{aligned}$$

Note the relation of entropy:

$$\begin{aligned} \mathcal{E}(F_0) &\geq \int_{\mathbb{T}^3 \times \mathbb{R}^3} \frac{1}{4} |f(t,x,v)| \mathbf{1}_{|f(t,x,v)|>\mu(v)} dv dx \\ &\quad + \int_{\mathbb{T}^3 \times \mathbb{R}^3} \frac{1}{4\mu(v)} |f(t,x,v)|^2 \mathbf{1}_{|f(t,x,v)|\leq\mu(v)} dv dx. \end{aligned}$$

Sketch of Proof of Step 1

Thus we have

$$III \leq CN^6(\lambda^{-2} + N^3) \left(\mathcal{E}(F_0) + N^{\frac{3}{2}} \sqrt{\mathcal{E}(F_0)} \right).$$

Result:

$$\begin{aligned} I + II + III &\leq C_q N^5 (1 - e^{-\lambda}) \overline{M} + \frac{C_q}{N^{q-10}} \overline{M} \\ &\quad + CN^6(\lambda^{-2} + N^3) \left(\mathcal{E}(F_0) + N^{\frac{3}{2}} \sqrt{\mathcal{E}(F_0)} \right). \end{aligned}$$

Sketch of Proof (Step 2)

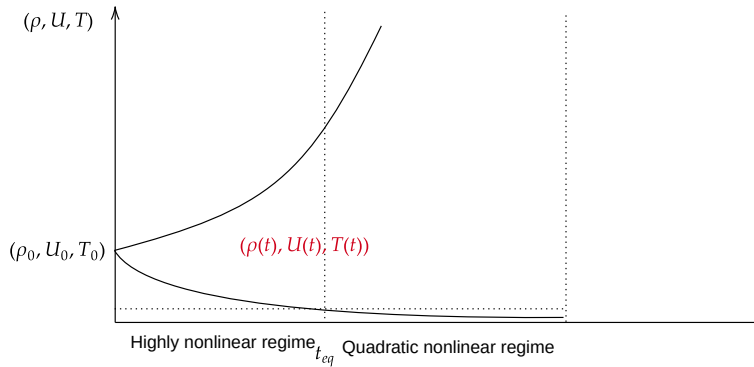


Figure: Step 2

Sketch of Proof (Step 2)

(Step 2) Estimate growth until t_{eq} .

Upper bound of the macroscopic fields:

$$(1) \quad C_0 e^{-C_q M_0^a t} \leq \rho(t, x) \leq C_q M_0,$$

$$(2) \quad |U(t, x)| \leq C_q M_0 e^{C_q M_0^a t},$$

$$(3) \quad C M_0^{-\frac{2}{3}} e^{-\frac{4}{3} C_q M_0^a t} \leq T(t, x) \leq C_q M_0 e^{C_q M_0^a t}.$$

We can reduce the nonlinear term to quadratic nonlinearity.

Estimate of the nonlinear term:

$$\|\Gamma(f)(t, x, v)\|_{L_{x,v}^{\infty,q}} \leq C_q M_0^n e^{C_q M_0^a t} \|f(t)\|_{L_{x,v}^{\infty,q}} \int_{\mathbb{R}^3} |f(u)|(1 + |u|^2) du.$$

Sketch of Proof (Step 3)

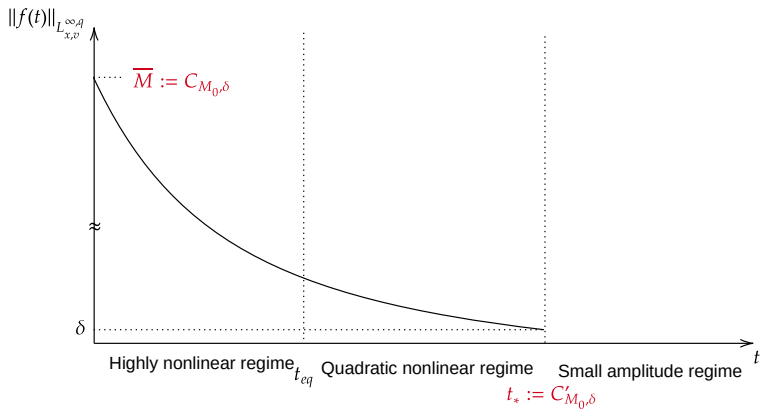


Figure: Step 3

Sketch of Proof (Step 3)

(Step 3) L^∞ estimate

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \leq C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} e^{-t/2} \left(1 + \int_0^t \|f(s)\|_{L_{x,v}^{\infty,q}} ds \right) + D.$$

We have

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \leq C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} \exp \left\{ 2 C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} \right\} (1 + Dt) e^{-t/2} + D.$$

Choose $\overline{M}$:

$$\overline{M} := 4 C_q M_0^{n+1} \exp \left\{ C_q M_0^a t_{eq} + 2 C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} \right\}.$$

We can extend the local in time unique solution until t_* .

Sketch of Proof (Step 4)

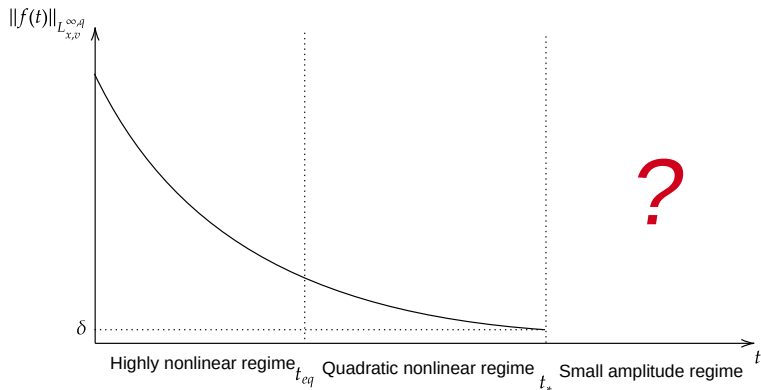


Figure: Step 4

Sketch of Proof (Step 4)

Lemma

There exists $\delta > 0$ such that if $\|f_0\|_{L_{x,v}^{\infty,q}} \leq \delta$, then there exists a unique global solution of the BGK model. Moreover, the following holds:

- ① *The solution is non-negative: $F(t, x, v) = \mu(v) + f(t, x, v) \geq 0$.*
- ② *The perturbation decays exponentially:*

$$\|f(t)\|_{L_{x,v}^{\infty,q}} \leq C_q \delta e^{-kt}.$$

for positive constants k and C_q .

Split the system:

$$\begin{aligned} \partial_t f_1 + v \cdot \nabla_x f_1 + f_1 &= \Gamma(f_1 + f_2), & f_1(0, x, v) &= f_0(x, v), \\ \partial_t f_2 + v \cdot \nabla_x f_2 &= (\mathbf{P}f_2 - f_2) + \mathbf{P}f_1, & f_2(0, x, v) &= 0, \end{aligned}$$

Summary and future works

- We found t_{eq} such that $|(\rho, U, T) - (1, 0, 1)| \leq \delta$ for $t \in [t_{eq}, \infty]$ under a priori assumption.
- We can measure the amount of increase until the finite time t_{eq} .

Thank you for attention!